

Request for criticism:
if ΔKL isn't innovation, what is it —
and how do we isolate the part that is?

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May 2026

What I am asking you to do. Attack the construct validity of this body of work, but constructively. I am not asking whether the regressions are coded correctly (assume they are; the code is reproducible) or whether the prose is polished. The defensive question — “is ΔKL a valid innovation measure?” — is a dead end, because I already suspect the honest answer is “not in the strong sense.” The two questions worth your time are sharper:

Q1. If ΔKL is not innovation, what is it? Give the construct its correct name, so the downstream results mean something precise rather than something inflated.

Q2. How do we isolate the part of ΔKL that genuinely is innovation? Treat the observed signal as a composite and tell me which decomposition cleanly extracts the innovation component — or prove no such component is identifiable from this data.

My own answer to Q1, which I want you to attack: ΔKL measures **lexical timing** — whether a filing’s registration language sat *ahead of* (positive) or *behind* (negative) its commercial category’s vocabulary trajectory at the moment of filing. It is not lexical novelty as such: a filing can be highly distinctive (high on both component KLS) yet have $\Delta KL \approx 0$ if it is equally far from the class’s past and future. Positive ΔKL means “wrote in language the field moved toward.” That construct is well-defined and, I think, defensible [**high**]. The open question is *why* a filing is ahead of the curve: firm foresight (the innovation interpretation), attorney drafting fashion, or mere exposure to a churning category. Sections 2–4 lay out the steelman for “it’s mostly the latter two,” my evidence, and the decomposition program for Q2.

1 The claims, briefly

There are two bodies of work sharing one measurement apparatus.

(A) The ΔKL papers (main.pdf, short.pdf). For each of 10,060,073 “clean” US trademark filings (1990–2021 cohorts), I compute a Kullback–Leibler surprise difference $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ on the goods/services description text, against per-day exponentially decay-weighted reference distributions of the same NICE class (half-life $H = 2y$). $\Delta KL > 0$ when a filing’s vocabulary is unusual against the recent past and close to the near future — the directional resonance signal of Murdock, Allen, & DeDeo (2017), applied for the first time to commercially-incentivised text. The papers claim: ΔKL is an innovation measure; it predicts post-registration survival (*U*-shaped) and

financial outcomes; it is statistically independent of patent counts ($r = +0.01$ on 174,569 firm-years) and of comparable correlation magnitude; the innovation tail aligns with the empirical content of optimal distinctiveness and the harvest tail is an unexplained surprise.

(B) The dynamism / ethnic-cluster work (dynamism.pdf, ethnic_clusters_note.pdf). Treating a firm’s first-ever filing (“debut”) as a proxy for business formation, I show trademark debuts cover 8–18% of Census High-Propensity Business Applications; I impute applicant race/ethnicity from surnames; I document ethnic-cluster industries, a Black trademark-rise / patent-fall divergence, and the collapse of an apparent “patent guild” signal once foreign inventors are stripped.

Everything in (B) inherits the validity of the measurement in (A), plus its own surname-imputation problems. So the load-bearing question is shared.

2 Steelman: the case that this measures nothing real

Here is the argument I most want you to either complete or refute. I find it uncomfortably strong. [moderate] that at least the first three links are serious and not fully answered by the current drafts.

1. **A trademark is a brand, not an invention, and the goods/services text is a legal instrument, not a description of novelty.** The g/s description exists to define the *scope of protection*, drafted by an attorney (or selected from the USPTO’s pre-approved *ID Manual*) to be as broad as the examiner will allow. Two firms selling physically identical products can write wildly different descriptions; a firm with a genuinely novel product can file boilerplate. So ΔKL measures the novelty of the *prose*, and the prose is several steps removed from the product. Why should I believe the prose tracks the innovation?
2. **Vocabulary novelty has at least four generators, and I have separated none of them.** A high- ΔKL filing uses words that are unusual-then-common. That could be (a) a genuine new product category (crypto, vape, NFT); (b) legal-drafting fashion; (c) attorneys copying each other’s successful protective language; (d) the USPTO updating its ID Manual, which mechanically shifts the reference vocabulary for everyone. The cannabis/crypto examples I parade are consistent with (a), but they are also exactly the cases where new ID-Manual entries appeared. I have not shown the signal is (a) rather than (d).
3. **The resonance interpretation was validated on a different species of text.** Murdock, Allen, & DeDeo (2017, *Cognition*) developed the framework on Darwin’s private reading notebooks; Barron, Huang, Spang, & DeDeo (2018, *PNAS*) on French Revolution parliamentary debates — in both, the text is a window into the author’s cognitive or rhetorical state, authored by the person whose novelty is being measured. A trademark g/s description is drafted for protection scope by a third-party agent. Importing the “resonance = the field moved toward this filing” interpretation assumes the signal means the same thing in a corpus where the author is a lawyer optimising a legal outcome. I assert the analogy; I do not re-validate it.
4. **Selection: I only ever observe the ~ 8 –18% of business starts that trademark.** Coverage against Census High-Propensity Business Applications is 8% (2004–08) rising to 18% (2019–24). The trademark-filing population is selected on branding intensity, consumer-facing orientation, and capitalisation. Every sentence I write “about firms” is really about the trademark-filing subset, and that subset is exactly the set of firms for which branding-vocabulary is salient — which is precisely the variable ΔKL measures. The selection is on the regressor.

5. **“Survival” is trademark maintenance, not firm survival.** My headline outcome — 5-year survival — is the owner paying the § 8 declaration-of-use fee. A live firm can abandon a mark; a dead firm’s mark can sit registered until renewal lapses. The outcome is a portfolio-management decision correlated with, but not identical to, firm viability. The U -shape on ΔKL could be a U -shape in *which marks firms bother to maintain*, not in which firms survive.
6. **The financial validation is small and one of the numbers is implausible.** The patent R^2 decomposition is on 8,118 firm-years; the gross-margin effect is +3.2pp per σ (SE 0.6); these are real but modest. The claimed +28 percentage-point 4-year excess return per σ of debut ΔKL is, on its face, [low] confidence and probably too large to believe — a σ move in a text statistic should not move 4-year returns by 28pp; I suspect a specification or sample-composition artefact and want it stress-tested.
7. **Independence from patents cuts both ways.** I sell $r = +0.01$ with patent counts as evidence that ΔKL is “a distinct innovation channel.” The skeptic’s reading is the opposite: ΔKL is uncorrelated with the one measure we *know* tracks invention because ΔKL is not measuring invention at all — it is measuring marketing/branding activity, which is orthogonal to R&D by construction. Independence is necessary for “distinct channel” but equally consistent with “not an innovation measure.”
8. **The ethnic / dynamism layer multiplies the problem.** Surname imputation is probabilistic, restricted to the $\sim 32\%$ of filers parsing as individuals, static to the 2010 Census, and weakest exactly for Black surnames (Smith/Johnson/Williams load on both White and Black). On top of that it inherits the whole proxy-validity question: if trademark debut is not business formation, the coverage ratios and ethnic-composition trends describe “who files trademarks,” not “who starts businesses.”

3 What I actually have for validity (honest accounting)

I am not claiming the steelman wins. Here is the evidence on the other side, with its limits stated.

- **Patent independence + comparable magnitude.** $r = +0.01$ with patent counts (174,569 firm-years); incremental 4-year-return R^2 of +0.0046 for ΔKL vs +0.0024 for $\log(1 + \text{patents})$, roughly additive. *Limit:* small R^2 absolute; comparable-to-patents is a low bar if patents themselves explain little return variance in this panel.
- **Survival is U -shaped under multiple operationalisations and in 32/44 NICE classes.** *Limit:* all of them are maintenance-based; the robustness is across definitions of maintenance, not across independent firm-outcome data.
- **Cluster-viability mediation absorbs 84% of the quadratic.** This is the one result that looks mechanism-like rather than mechanical. *Limit:* the mediator is itself built from trademark data (cosine-neighbour survival), so it does not break out of the corpus.
- **Coverage brackets Dinlersoz et al. (2018).** Our 8–18% spans their restricted-LBD 10–15%. *Limit:* agreement on the *coverage fraction* is not validation that the covered filings *mean* anything; it just says we count debut filers the way Census counts them.
- **Face validity.** Canonical winners skew positive ΔKL ; flux-neutral tokens (organisational, conformity) die. *Limit:* selected ex-post; not a test.

My own read [**moderate**]: I can defend “ ΔKL is a nontrivial, patent-orthogonal signal that predicts trademark-level outcomes and correlates weakly with firm financials.” I cannot currently defend “ ΔKL is an innovation measure” against a determined skeptic, because every outcome I validate against is either inside the trademark corpus or a small noisy financial panel.

4 Q1: if not innovation, what is it? — four candidate names

A ladder of constructs from strong to weak. Tell me which is the correct name for what ΔKL measures.

(I) Strong — “ ΔKL is an innovation measure.” The current framing. Requires that vocabulary novelty in g/s prose tracks product/market innovation. My confidence this survives review as stated: [**low**].

(II) Medium — “ ΔKL is a leading indicator of commercial category emergence.” Claims the corpus-level signal (which *vocabularies* are forming and being adopted) is real and useful, while staying agnostic about whether any individual filer is “innovative.” The cannabis/crypto/NFT waves are the evidence; the firm-level survival/returns become secondary. Confidence: [**moderate**].

(III) Weak/descriptive — “ ΔKL measures the linguistic position of a filing relative to its NICE class.” Purely descriptive. Every downstream correlation (survival, returns, ethnic clusters) becomes an empirical regularity to be explained, not a validation of an innovation construct. This is almost certainly defensible [**high**] but gives up the headline.

(IV) Activity proxy — “a trademark debut is formal commercial activity with branding intent.” For the dynamism work specifically: not innovation, not business formation *per se*, but a measurable, name-disclosed, near-real-time signal of branding-intent commercial activity, explicitly the 8–18% branding-intensive slice of formation. Confidence: [**moderate**].

My current bet [**moderate**]: the correct primitive name is (III) lexical timing; (II) is the defensible *corpus-level* claim built on top of it; (I) is only licensed for the subset of ΔKL that survives the isolation program in the next section; and (IV) is what the dynamism work should stand on, needing none of the above. The restructure I am tempted by: lead with (III) as the measurement contribution, establish (II) with the category-emergence evidence, present firm-level survival/financial results as *regularities under (III) awaiting the (I) decomposition*, and let the dynamism work rest on (IV). Tell me if that is rigor or cowardice.

5 Q2: how do we isolate the innovation component?

Model the observed statistic as a composite:

$$\Delta KL_{\text{filing}} = \underbrace{\beta (\text{firm innovation})}_{\text{the part we want}} + \underbrace{\gamma (\text{attorney/drafting style})}_{\text{fashion}} + \underbrace{\delta (\text{category churn exposure})}_{\text{being in a hot class}} + \varepsilon.$$

The whole question is whether β (firm innovation) is separately identifiable. Five tests, ranked by decisiveness $\times$ feasibility. Tell me which is decisive, which is hopeless, and what I am missing.

(C) Within-firm ΔKL vs. patent timing — *runnable now*. For firms that both patent and trademark, test whether a firm’s high- ΔKL trademark years coincide with its high-patent years. Within-firm temporal co-movement is hard to explain by drafting fashion or category churn and would be direct evidence $\beta > 0$. Temporal independence would say ΔKL and invention are unrelated *even inside the same firm*. We have `firm.dkl-vs.patents.parquet`; I can run this immediately.

(D) Term-provenance decomposition — *runnable now*. Split each filing’s ΔKL contribution into tokens *new to the class* vs. recombined pre-existing tokens. Genuine novelty should load on new tokens; drafting fashion on unusual recombination of old tokens. If the survival/return effects concentrate in the new-token component, that is the innovation part isolated. We have the per-class vocab and per-filing tokens.

(A) Attorney / correspondent fixed effects — *cheap re-parse*. The strongest design. The USPTO XML carries the correspondent/attorney of record; we already re-parsed once for applicant country, so adding the attorney field is incremental. Within-attorney variation in ΔKL nets out γ (drafting style) by construction. If the survival/financial effects survive attorney fixed effects, ΔKL is not drafting fashion.

(B) ID-Manual-only vs. free-text split — *moderate*. The USPTO *ID Manual* is a controlled vocabulary of pre-approved descriptions. Filings using only ID-Manual language have mechanically constrained vocabulary (ΔKL is pure δ , category-churn exposure); filings with free-text descriptions carry discretionary language that can encode β . If the effects concentrate in free-text filings, that localises the innovation component. Needs the ID Manual (downloadable).

(E) ID-Manual adoption prediction — *moderate*. The institutional validation test: does a high- ΔKL filing use terms that the ID Manual *later* codifies? If filings that are “ahead of the curve” by ΔKL are using terms the USPTO itself subsequently standardises, the leadership is real and externally validated, not an artefact of my reference window. Needs ID-Manual version history.

My plan, pending your verdict: run (C) and (D) now, add the attorney field on the next re-parse for (A), and treat (B)/(E) as the ID-Manual work package. If (C) shows within-firm co-movement and the (D) new-token component carries the survival effect, I will claim (I) *for that isolated component only* and keep (III) as the name for the rest. If both come back null, (I) is dead and I retreat to (III)+(II)+(IV).

6 Specific questions

1. Is (A) attorney fixed effects actually clean, or does the firm choose its attorney partly *for* drafting style, so that attorney FE absorbs real firm-level signal along with the fashion?
2. Is (C) within-firm ΔKL /patent co-movement convincing evidence of $\beta > 0$, or is there a confound (e.g., both spike when the firm enters a new product line) that co-movement cannot rule out?
3. For (D): is “new-to-class token” the right operationalisation of the innovation component, or should novelty be measured against the whole-corpus vocabulary rather than the class?
4. Is there an external, non-trademark, non-financial outcome to validate survival against — business-registration death, UCC-lien termination, Form-D withdrawal, domain expiry — that breaks the maintenance-vs-survival circularity?

5. Does the $+28\text{pp}/\sigma$ return number kill my credibility on sight? Drop it pending re-estimation?
6. For the ethnic work: is surname imputation usable given the Black-surname cross-loading (Smith/Johnson/Williams), or do I restrict to the discriminating Hispanic and API analyses and demote Black claims to a flagged-caveat appendix?

7 First isolation results: lead-lag, cross-class transfer, case studies

Three of the Q2 tests have first results. They land on a sharper picture than either the strong (I) or full-retreat (III) framing anticipated. The headline: ΔKL tracks invention activity at the *between-firm* level but not (cleanly) within-firm at annual resolution, and an idea-transfer signature is visible in the filing-level vocabulary data.

Lead-lag: patents *do* lead high- ΔKL trademarks between firms, but not within firms at this resolution

The pipeline framing (patents $\rightarrow$ products $\rightarrow$ trademarks) predicts that contemporaneous patent- ΔKL correlation can be zero while lagged correlation is positive. We rebuilt an outer-join firm $\times$ year panel restricted to patent-active firms ($n = 54,323$ firms with ≥ 1 patent in any year), keyed patents to *application* date (the invention-timing proxy) rather than grant date, and computed $\text{corr}(\text{patents}(t), Y(t+k))$ for $k = -5 \dots +5$, where Y is either the firm's mean ΔKL or its count of positive- ΔKL filings that year.

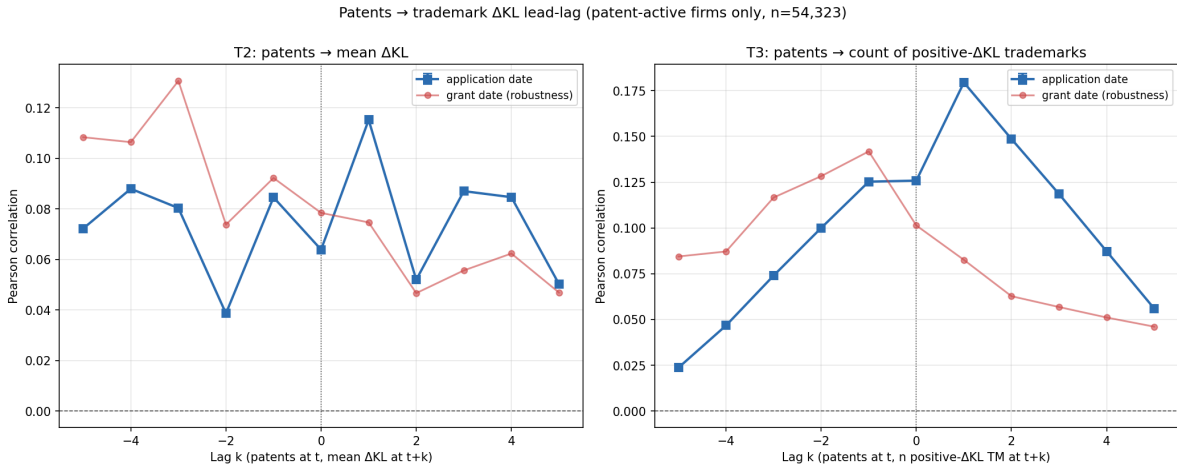


Figure 1: Lead-lag cross-correlation. Left: patents $\rightarrow$ mean ΔKL . Right: patents $\rightarrow$ count of positive- ΔKL trademark filings. Blue = application date (clean), red = grant date (robustness; grant lags application by $\sim 2.5\text{y}$ so trademarks can legitimately precede grant). 95% CIs via Fisher z .

The compositional caveat. Within-firm FE regressions of the same outcomes on patent-application lags 0–3, clustered SE, restricted to firms with ≥ 5 firm-years of trademark ΔKL ($n = 1,146$ firms, 13,425 firm-years), produce *statistically insignificant* coefficients across all lags (largest $|t| \approx 1.7$, all $p > 0.08$). Direction on positive- ΔKL count is consistent with the pipeline (lag-1 coef $+0.014$, lag-2 coef $+0.014$) but neither passes significance.

k	n	$r(\text{pat}_{\text{app}}, \overline{\Delta KL})$	$r(\text{pat}_{\text{app}}, n_{\Delta KL > 0})$	$r(\text{pat}_{\text{grant}}, n_{\Delta KL > 0})$
-3	4,811,278	+0.080	+0.074	+0.117
-2	5,362,199	+0.039	+0.100	+0.128
-1	4,622,243	+0.085	+0.125	+0.142
0	5,035,558	+0.064	+0.126	+0.101
+1	4,590,147	+0.115	+0.179	+0.083
+2	3,658,631	+0.052	+0.149	+0.063
+3	5,170,367	+0.087	+0.119	+0.057
+5	2,825,536	+0.050	+0.056	+0.046

Table 1: Pearson correlation between firm-year patents and a future trademark outcome. Application-date $\overline{\Delta KL}$ peaks at $k = +1$ ($r = +0.115$); positive- ΔKL count peaks at $k = +1$ ($r = +0.179$). Grant-date positive- ΔKL count peaks at $k = -1$, exactly what the $\sim 2.5\text{y}$ application-to-grant lag predicts if trademarks come during patent pendency. n is firm-year-pair observations.

What this means [**high**]: the population-level lead-lag is *between-firm*. Patent-active firms file systematically more positive- ΔKL trademarks than non-patent firms, and the between-firm covariance accumulates across years to produce the $r = +0.18$ at $k = +1$. *Within* a firm, knowing how many patents it filed last year does not predict how many positive- ΔKL trademarks it will file this year, at annual resolution and on this sample.

Interpretation for Q1 and Q2. The straight pipeline narrative (“last year’s patent caused this year’s trademark”) is not supported as a within-firm temporal mechanism in our data. What *is* supported is firm-level co-selection: the population of patent-active firms overlaps with the population of high- ΔKL trademark filers, and the lead-lag profile of the population covariance is consistent with the broad direction of the pipeline (patents earlier, trademarks later). The conservative reformulation: ΔKL -positive trademarks identify the same set of firms that patenting identifies, with a roughly +1-year lag at the population level, but ΔKL does *not* cleanly time-resolve a within-firm innovation pulse. This is closer to reframing (I) being partially licensed — for the patent-overlap subset, ΔKL does co-localise with invention activity — than to either (III) being the full retreat or (I) being licensed in the strong causal sense.

Reasons the within-firm test may underpower a real pipeline. $n = 1,146$ firms is small; annual resolution may be too coarse (patent-to-launch cycles are often months, not years); patents come in bunches; not every patent leads to a trademark; and the relationship is plausibly heterogeneous across industries (pharma vs. software vs. hardware) which we have not stratified. Higher-frequency timing data (monthly patent applications $\rightarrow$ monthly trademark filings) and within-CPC-to-NICE stratification would be the next steps.

Cross-class vocabulary transfer: positive ΔKL comes at the cost of registration

A separate test of what ΔKL measures: identify trademark filings whose goods/services prose imports tokens recently ascendant in a *different* NICE class (vocabulary moving across industry boundaries). On 7,290,115 clean-panel filings, with the operational definition “a filing imports ≥ 2 tokens currently rising in some other class but not yet rising in its own,” transfer filings have **higher ΔKL** (+0.033 pooled, +0.031 within class $\times$ year cell) and *simultaneously* **lower registration** rate (−4.4pp pooled, −2.8pp within cell, both $z > 11$). The face-validity

examples are correct: the top-intensity transfer filings are 2014–2015 vape and e-cigarette filings importing class-34 (tobacco) vocabulary into classes 1, 10, 11, 35, 38.

This is the cleanest result on Q1 so far. Cross-class imports both *look novel* (positive ΔKL , because they carry tokens that are unusual against the destination class’s recent past) and *fare worse at the examination gate* (lower registration, because examiners and prior-art collisions disfavour them). Both priors I considered turn out to be partially right *at the same time*: validated-novelty and category-mismatch are simultaneous forces, not alternatives. Detail: see `paper/cross_class_transfer_note.pdf`.

Firm case studies: the pipeline is visible firm-by-firm even when the panel won’t show it

A complementary qualitative check on eight IP-heavy firms (Apple, Tesla, NVIDIA, Pfizer, Moderna, Dyson, Amazon, Salesforce). The cleanest patent→trademark linkages, where the within-firm pipeline is visible by inspection:

- **NVIDIA TEGRA** (2008, $\Delta KL = +0.42$), **DGX** (2017, $+0.46$), **RTX** (2018, $+0.28$) — three textbook technology discontinuities (mobile SoCs, AI compute, ray tracing), each landing a $> +0.25 \Delta KL$ on the firm’s own trademark filing at the time of the corresponding patent-cluster activity.
- **Tesla 2015 firm-mean** $\Delta KL = +0.33$, $n = 12$ **filings**, the firm’s largest single-year ΔKL spike, coincident with the Powerwall and Powerpack stationary-storage launches.
- **Amazon KINDLE** (2006, $+0.19$), **ECHO** (2014, $+0.16$), **PRIME AIR** (2014, $+0.15$) — three sequential novel-vocabulary filings tracking Amazon’s hardware push.

Honest negative findings: Pfizer’s pipeline is broken by IP-holdco filings (Warner-Lambert / Wyeth / BioNTech as legal owners); Apple IPHONE (1996) is a parked filing 11 years pre-launch, unrelated to the eventual product; Salesforce SLACK is acquisition-conditional. Detail: `paper/firm_case_studies_note.pdf`.

What the case studies and the panel jointly say. For the right kind of firm (hardware-with-a-clear-new-product-name and not a holdco) the within-firm pipeline is visible by eye — NVIDIA is near-perfect. The panel regression’s within-firm null is therefore *not* “no firm exhibits the pipeline,” it is “averaging across 1,146 firms with ≥ 5 TM- ΔKL -years does not recover an annual-resolution lag coefficient.” Both are honest. The question for you (reviewer): is the right test stratified by firm type (hardware vs. pharma vs. software) and at higher temporal resolution, or is the population-level lead-lag + case-study qualitative evidence enough?

8 Artifacts and reproduction

All on branch `business-dynamism` of the project repo. `main.pdf` (long ΔKL paper, 223pp), `short.pdf` (7pp), `dynamism.pdf` (13pp), `ethnic_clusters_note.pdf` (21pp). Pipelines: `Makefile` and `Makefile.dynamism`. Every figure regenerates from `scripts/` against the per-NICE-class parquets; the surname work uses the 2010 Census surname file and a full re-parse of the USPTO XML for applicant country (`owner_address.parquet`, 13,993,768 records).

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